MD REHAN ALAM

CS Undergraduate (AI/ML Focus)

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PROFESSIONAL SUMMARY

AI and Machine Learning-focused CSE student with strong foundations in Python and web development. Combines analytical problem-solving with digital media expertise to build technical solutions and drive online audience engagement.

PROJECTS

KAVACHAM || GitHub
Role: Frontend Developer
Detected outbound personal-data exposure, inspected third-party APK risk, preserved encrypted evidence, prepared erasure requests, and organized a reviewable escalation timeline.
Technologies / Tools Used: *React, Node.js, Tailwind CSS*

HYBRID-RAG SEARCH ENGINE || GitHub
Role: Solo Developer
Developed an enterprise-grade hybrid **RAG application** for precision technical search, utilizing **dual lexical-dense indexing** and **cross-encoder reranking** to generate hallucination-free answers.
Technologies / Tools Used: *python, FastApi, Generative AI*

FORENSIC SHIELD || GitHub
Role: Frontend Developer
Developed an offline multimodal deepfake detection engine that analyzes spatial facial boundary artifacts, audio spectral phase patterns, and phoneme-viseme lip-sync alignment to generate tamper-proof, **SHA-256 verified PDF evidence reports** on consumer hardware using **Python, PyTorch, and INT8 ONNX Runtime**.
Technologies / Tools Used: *React, Node.js, Tailwind CSS*

CineMind AI || GitHub
Role: Solo Developer
Developed an **AI-powered content-based movie recommender** using FastAPI and Vanilla JS. Utilized NLP techniques, including **CountVectorizer** and **PorterStemmer**, along with Cosine Similarity to compute real-time vector distances across **4,800+ films**. Designed a Glassmorphism UI and implemented a LocalStorage watchlist.
Technologies / Tools Used: *NLP, Flask, Tailwind CSS*

Air-Cursor || GitHub
Role: Solo Developer
Developed a real-time, gesture-controlled interactive web portfolio using **Python and Vanilla JavaScript**, leveraging **MediaPipe and WebSockets** for hands-free spatial navigation and seamless frontend interactions without a physical mouse.
Technologies / Tools Used: *OpenCV, mediapipe, Node.js*

SKILLS

Databases: MySQL, Firebase, Supabase, MongoDB
Frameworks & Libraries: React.js, Next.js, React, Skit-learn, TensorFlow, Keras, PyTorch
Programming Languages: Python, C, Java, HTML, CSS, TypeScript, JavaScript
Languages: Hindi, English
Technical Skills: OOPs, Data Structures and Algorithms, Machine Learning (ML)
Tools & Platforms: Git, GitHub, VS Code, Antigravity

AWARDS & ACHIEVEMENTS

Internal Smart India Hackathon 2026 Winners (Campus)
Swami Vivekananda University
Co-developed an **edge-compatible multi-modal forensic detection engine** for cyber law enforcement, digital forensics units, and judicial verification.

Citadel 1.0 National Finalists
University of Engineering & Management, Kolkata
Developed **Kavacham (The Defender)**, a local-first privacy defense and DPDP evidence-support platform for automated compliance and secure audits.